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Unit 3 : Session 2 : SLO 1 — Half Adder and Full Adder

■ 1. Label the Components (Remembering)

- Input 1: A Input 2: B
- Output 1: Sum Output 2: Carry
- Gate 1: XOR gate Gate 2: AND gate

■ 2. Complete the Truth Table — Full Adder (Understanding)

- $A=0, B=0, C_{in}=0 \rightarrow S=0, C_{out}=0$
- $A=0, B=0, C_{in}=1 \rightarrow S=1, C_{out}=0$
- $A=0, B=1, C_{in}=0 \rightarrow S=1, C_{out}=0$
- $A=0, B=1, C_{in}=1 \rightarrow S=0, C_{out}=1$
- $A=1, B=0, C_{in}=0 \rightarrow S=1, C_{out}=0$
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- $A=1, B=1, C_{in}=1 \rightarrow S=1, C_{out}=1$

■ 3. Trace the Circuit (Applying): $A = 1, B = 0, C_{in} = 1$

- Output of first XOR gate: 1 ($1 \text{ XOR } 0 = 1$)
- Output of first AND gate: 0 ($1 \text{ AND } 0 = 0$)
- Final S (Sum) output: 0 ($1 \text{ XOR } 1 = 0$)
- Output of second AND gate: 1 ($1 \text{ AND } 1 = 1$)
- Final Cout (Carry Out) output: 1 ($1 \text{ OR } 0 = 1$)

■ 4. Compare and Contrast (Analyzing)

- Number of inputs — Half Adder: 2 | Full Adder: 3
- Names of inputs — Half Adder: A, B | Full Adder: A, B, C_{in} (carry in)
- Number of outputs — Half Adder: 2 | Full Adder: 2
- Names of outputs — Half Adder: Sum, Carry | Full Adder: Sum, Cout (carry out)
- Accepts carry from previous stage — Half Adder: No | Full Adder: Yes
- Primary purpose — Half Adder: adds two single bits | Full Adder: adds three single bits including a carry-in

■ 5. Design a Multi-Bit Adder (Creating) — 2-bit ripple-carry adder

- Stage 1 (LSB): one HALF ADDER with inputs A_0 and $B_0 \rightarrow$ output $S_0 = A_0 \text{ XOR } B_0$, carry $C_1 = A_0 \text{ AND } B_0$.
- Stage 2 (MSB): one FULL ADDER with inputs A_1, B_1 and $C_{in} = C_1 \rightarrow$ output S_1 and final Cout.
- Connections: $A_0, B_0 \rightarrow$ HA (S_0, C_1); $C_1, A_1, B_1 \rightarrow$ FA (S_1, Cout).
- Outputs: S_0 (bit 0), S_1 (bit 1), Cout (final carry).